

FOODGRID AI

An AI-Assisted Framework for Surplus Food Redistribution, Demand Matching, and Sustainable Logistics

Research Paper — Draft 1

Evidence foundation, proposed architecture, research questions, and experimental plan

Research status: FoodGrid AI is presented as a proposed system/prototype. Future performance figures must be reported as measured experimental results or clearly labeled simulations.

Evidence point	Current evidence
Food wasted in 2022	1.05 billion tonnes globally
Share of food available to consumers wasted	About one-fifth
Estimated global GHG contribution of FLW	8-10%
UN SDG target	Halve per-capita global food waste at retail/consumer levels by 2030

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Abstract

Food loss and waste (FLW) is a major sustainability, food-security, and resource-efficiency challenge. UNEP reported that approximately 1.05 billion tonnes of food were wasted in 2022 at the retail, food-service, and household levels. FAO describes recovery and redistribution of safe and nutritious surplus food as an established intervention involving food banks, community kitchens, shelters, and other organizations. FoodGrid AI proposes a digital coordination framework that connects food donors, recipient organizations, and logistics providers through data-assisted matching and routing. The proposed architecture includes surplus registration, food classification, eligibility and shelf-life assessment, demand matching, route optimization, delivery tracking, and impact reporting. The research objective is not to claim that AI alone solves hunger, but to test whether better coordination can reduce matching time, unnecessary transport, and avoidable loss of edible surplus. The first research phase will establish a literature-backed architecture and a reproducible simulation against a baseline manual-matching process.

Keywords: food waste; food redistribution; artificial intelligence; logistics optimization; food security; sustainability; demand matching; route optimization

1. Introduction

Food waste is both a global environmental issue and a coordination problem. UNEP's Food Waste Index 2024 estimated that 1.05 billion tonnes of food were wasted in 2022, approximately one-fifth of food available to consumers. UNEP also reported that food loss and waste is associated with 8–10% of global greenhouse-gas emissions and significant use of land and freshwater. These figures demonstrate the scale of the problem, but they do not by themselves establish that any single digital intervention will solve it.

FAO distinguishes food loss from food waste and emphasizes that safe recovery and redistribution can make productive use of surplus food. FAO notes that redistribution operates through food banks, social supermarkets, food pantries, soup kitchens, shelters, and community or charitable programs. This supports the central premise of FoodGrid AI: a useful technology layer should connect existing actors rather than assume one platform can replace the entire food system.

The proposed research therefore focuses on a narrower and testable question: whether an AI-assisted coordination system can improve the speed and efficiency with which surplus food is matched to eligible recipients while respecting food-safety and logistics constraints.

2. Problem Statement

Surplus food can become unavailable for human consumption because donor staff may not have enough time to identify recipients, recipient organizations may lack timely information about available supply, and transport capacity may not be available when a short-shelf-life item becomes ready for collection. These are coordination failures rather than simply production failures.

FoodGrid AI addresses this coordination layer. The system is intended to represent supply, demand, time constraints, location, capacity, and transportation requirements in one operational model. The platform should recommend actions; it should not independently certify food safety or override legal requirements.

3. Research Objectives

- Develop a research-backed architecture for connecting surplus-food donors with eligible recipient organizations.

- Model food matching using quantity, food category, time-to-expiry, recipient demand, capacity, distance, and eligibility constraints.
- Evaluate route-selection methods that seek to reduce travel distance/time while meeting delivery constraints.
- Measure whether automated coordination can outperform a defined manual or nearest-neighbor baseline in simulation.
- Develop transparent impact metrics for rescued food, meals served, transport activity, and environmental indicators.
- Create a global research directory of potential ecosystem participants using verified public sources, while separating research examples from confirmed FoodGrid partners.

4. Research Questions

RQ1: Can automated supply-demand matching reduce the time required to allocate surplus food compared with a manual baseline?

RQ2: Can constrained route optimization reduce delivery distance or time while maintaining recipient and food-handling requirements?

RQ3: Which data fields are most important for matching short-shelf-life food to recipient demand?

RQ4: How sensitive are system results to uncertainty in quantity, shelf life, demand, traffic, and vehicle availability?

RQ5: What operational and regulatory conditions must be satisfied before a prototype can be tested with real organizations?

5. Proposed FoodGrid AI System Architecture

Stage	Input	Proposed function	Output
1. Surplus logging	Photo + donor data	Register quantity, category, time, location	Structured surplus record
2. Food assessment	Images + metadata	Classification and rule-based eligibility check	Food category + confidence + status
3. Shelf-life model	Food type + timestamp + handling data	Estimate remaining usable window	Time-to-expiry estimate
4. Demand matching	Supply + NGO demand + capacity	Score feasible recipient options	Ranked recipient matches
5. Logistics	Locations + vehicle constraints	Route/assignment optimization	Dispatch plan
6. Tracking	Pickup/delivery events	Record chain of custody and status	Delivery record
7. Impact reporting	Delivery + weight + destination data	Calculate defined indicators	Impact dashboard/report

6. AI and Optimization Components

Computer vision. A prototype can classify broad food categories from donor photographs. This is an assistance mechanism, not food-safety certification. Research should report accuracy, confidence, and failure cases.

Shelf-life prediction. A model may estimate remaining usable time from food type, timestamp, storage conditions, and other data. Because shelf life is safety-sensitive, the design should use conservative rules, human verification, and applicable regulations.

Demand matching. Candidate recipients can be ranked using food compatibility, requested quantity, capacity, geographic distance, urgency, and eligibility.

Route optimization. A vehicle-routing or assignment model can seek to minimize travel distance/time subject to vehicle capacity, pickup and delivery windows, and other constraints.

Impact calculation. Environmental calculations should use transparent, source-specific emission factors rather than a universal conversion such as '1 kg food = X kg CO₂e'.

7. Stakeholder Ecosystem

Stakeholder	System role	Research variable
Retailers / supermarkets	Report surplus and pickup windows	Quantity, category, frequency
Farms / producers	Report harvest or production surplus	Seasonality, quantity, location
Restaurants / kitchens	Report safe surplus food	Meal count, time window, handling
NGOs / food banks	Declare demand and receiving capacity	Demand, capacity, eligibility
Logistics providers	Provide transport capacity	Vehicle, capacity, availability, route
Municipal / public bodies	Policy and waste-diversion context	Targets, regulations, geography
FoodGrid AI	Coordinate information and optimization	Matching, routing, tracking, reporting

8. Experimental Design

The first experimental phase should use synthetic or publicly documented scenarios rather than claiming real-world performance. A dataset can contain donor locations, recipient locations, food categories, quantities, recipient demand, time windows, and vehicle capacities. Two systems can be compared: (A) a baseline such as nearest eligible recipient/manual-style matching and (B) the FoodGrid optimization approach.

Primary metrics: average matching time, percentage of surplus successfully assigned, delivery distance, delivery time, vehicle utilization, unmet recipient demand, and number of feasible deliveries. Secondary metrics can include estimated transport emissions and cost when assumptions and emission factors are documented.

Research integrity rule: performance figures such as '40% faster dispatch' must only appear after an actual experiment produces them. Until then, they belong in the hypothesis or target-performance section, not the results section.

9. Evidence-Based Global Context

UNEP's Food Waste Index 2024 provides global-scale evidence for the magnitude of food waste. FAO provides complementary definitions and guidance on food loss, food waste, and recovery/redistribution. The UN Sustainable Development Goals establish SDG 12.3, calling for halving per-capita global food waste at retail and consumer levels by 2030. These sources establish relevance; they do not validate FoodGrid AI's future performance.

The U.S. EPA's Wasted Food Scale places prevention, donation, and upcycling among the most preferred pathways for managing wasted food from an environmental perspective. This supports studying donation and redistribution within a broader waste-prevention hierarchy.

10. Limitations and Safety Requirements

- FoodGrid AI cannot determine food safety from a photograph alone.
- Shelf-life prediction is uncertain and must not replace applicable food-safety rules, labeling, storage requirements, or human verification.
- A global network must adapt to country-specific laws, tax rules, data-protection requirements, and food-donation regulations.
- A prototype organization database must distinguish verified partners from potential or research-only organizations.
- Environmental impact estimates depend on assumptions, food categories, transport modes, and system boundaries.
- Simulation results cannot be presented as evidence of real-world impact until validated through field deployment.

11. Proposed Global Organization Research Database

A later appendix will catalogue supermarkets, food producers, NGOs, food banks, logistics providers, and public organizations by country and city. Each record should include the official name, category, city/country, official website, publicly listed address where appropriate, relevant program information, and the source used for verification. Each organization will be labeled as 'verified FoodGrid partner', 'potential partner', or 'research reference'. No organization should be described as a partner without documented consent or agreement.

12. Conclusion

FoodGrid AI presents a plausible research direction at the intersection of food-waste reduction, food redistribution, artificial intelligence, and logistics optimization. The strongest scientific version is not a claim that software will eliminate hunger. Instead, it is a testable hypothesis that better information coordination and optimization can reduce friction in moving safe surplus food to organizations that can use it. The next stage should focus on literature review, dataset design, baseline definition, algorithm selection, simulation, and careful measurement.

References — Initial Evidence Set

- [1] United Nations Environment Programme (UNEP). Food Waste Index Report 2024 / Key Findings. <https://www.unep.org/resources/publication/food-waste-index-report-2024>
- [2] Food and Agriculture Organization of the United Nations (FAO). Recovery and redistribution of safe and nutritious food for human consumption. <https://www.fao.org/platform-food-loss-waste/food-waste/food-waste-reduction/recovery-and-redistribution-of-safe-and-nutritious-food-for-human-consumption/en>
- [3] FAO. Food Loss and Food Waste — Policy Support and Governance Gateway. <https://www.fao.org/policy-support/policy-themes/food-loss-and-food-waste/en>
- [4] United Nations Department of Economic and Social Affairs. Sustainable Development Goal 12. <https://sdgs.un.org/goals/goal12>
- [5] U.S. Environmental Protection Agency (EPA). Wasted Food Scale. <https://www.epa.gov/sustainable-management-food/wasted-food-scale>
- [6] U.S. EPA. From Field to Bin: The Environmental Impacts of U.S. Food Waste Management Pathways. <https://www.epa.gov/land-research/field-bin-environmental-impacts-us-food-waste-management-pathways>

Next draft: expand the literature review with peer-reviewed AI/logistics research, define the data schema, construct the global organization-source table, specify the optimization model, and add a reproducible simulation.